

Remote Work vs On-Site Wages

This research project aims to answer the question of whether remote work pays less than non-remote work, even with education factored in. To analyze this, data were drawn from IPUMS CPS and processed in R-studio to show the trends of remote and non-remote wages vs education level. The final result, on average, shows that remote work will pay more than its counterpart and even increases in difference the higher the education level.

“Do remote workers have lower wages, even if the worker has the same education?” is the primary research question for this project. The motivation behind this question was due to the recent rise of remote work due to the COVID pandemic. Naturally, the question became whether or not workers got paid more or less than those who did not work remotely, and if their education level played a factor in that. The dataset found from IPUMS specifically recorded the hourly wages of each worker and whether or not they did remote work, along with their education level. Answering the research question, I expect to find that those working remotely should be earning less than their physical/non-remote counterparts, as it is unlikely for employers to see more value in a virtual worker than a worker who is physically in front of them.

The data set was directly drawn from IPUMS CPS, with the variables YEAR, MONTH, HOURWAGE2, EDUC, and TELWRKPAY for simplicity, with further error filtering to remove NIU codes, with the timeline between 2022 month 10, to 2025 month 8. The data set variable HOURWAGE2 is the hourly wage of the worker in USD, EDUC is the education level recorded of the worker (2-73 PreK-12, 80-111 Bachelor, 120-123 Master's, 124-125 Doctorates), and TELWRKPAY is whether or not the worker does remote work. This data set is straightforward in its nature, with 195,500 observations post-error filtering. The reasoning for the strange times and the use of HOURWAGE2 is due to IPUMS CPS not recording TELWRKPAY until October 2022, and HOURWAGE1 records stopped in 2023 and its successor is HOURWAGE2. The very original data (unprocessed) contains 4,031,169 observations with a file size of 73.5 MB; this consistently crashed R-studio, and an overwhelming majority of it contains NIU codes. This led to the creation of the final cleaned data set with 195,500 observations at 4.6 MB (a 95% size reduction) that was usable.

There are a total of 4 graphs made, 2 of them being linear regression models to get a clear comparison. Graphs 1 and 2 demonstrate the Average Wage vs Education Level, total and separate remote/not-remote. Graphs 3 and 4 are the linear regression models studying the trends of 1 and 2, with Graph 4 specifically studying the average growth monthly separately on remote/not-remote over the years. The reason graphs 3 and 4 are linear regression models is to see the raw data volatility, while graphs 1 and 2 are for simplicity. The results from 1 and 2 demonstrate that wages dip/stagnate until college education, where they then rise with each

education level. However, graph 2 shows that remote work surpasses the wages of on-site wages at around 8th grade, with the regression on graph 3 proving this wage difference trend starting at around 6th grade. Graph 4 linear regression combines all education levels for remote/on-site and shows the average total wage per month over the years, increasing, showing that remote work is consistently higher than on-site, but also much more volatile, with on-site being consistently stable. In total, throughout the 4 graphs, the results reveal that online work *consistently pays more* than on-site as education increases after grade 8, and in fact, the differences only get bigger the higher the education level. The results, however, cannot reveal whether or not the remote work wage was higher due to social/job factors. There is a strong possibility that remote work wages are higher due to jobs that offer remote work being collar jobs (ei, banking, research positions, etc). If the experiment were to be run again, ideally, the best way to gauge this would be to specially run a consensus on white collar jobs in a particular job sector and ask the wages of those working in office vs those working remotely, thus removing the blue collar vs white collar and job sector variable.

Based on the graphs and from Data Table 1, the results show that my original hypothesis was wrong; on average, remote workers have *higher* wages, even if the worker has the same education. My conclusion then is that remote work, on average, pays more after grade 8 education. However, it is more than likely remote a higher wage because jobs that can offer remote work will often already be paying a higher wage vs those that don't usually offer a high wage. My project requires additional research on the matter to verify this assumption I am making, as verifying per job (and age) with the region would provide much clearer results.

In conclusion, remote work jobs, on average, pay more than on-site jobs after graduating 8th grade. Knowing this data, I intend to look into my own life career options and be more open to remote work, knowing now that remote work is not nearly as poor as once assumed.

Appendix:

```
# all education, average wage remote vs on-site
data |>
  mutate(online_work= TELWRKPAY == 1) |>
  group_by(online_work) |>
  summarise(wage_avg= mean(HOURWAGE2))
```

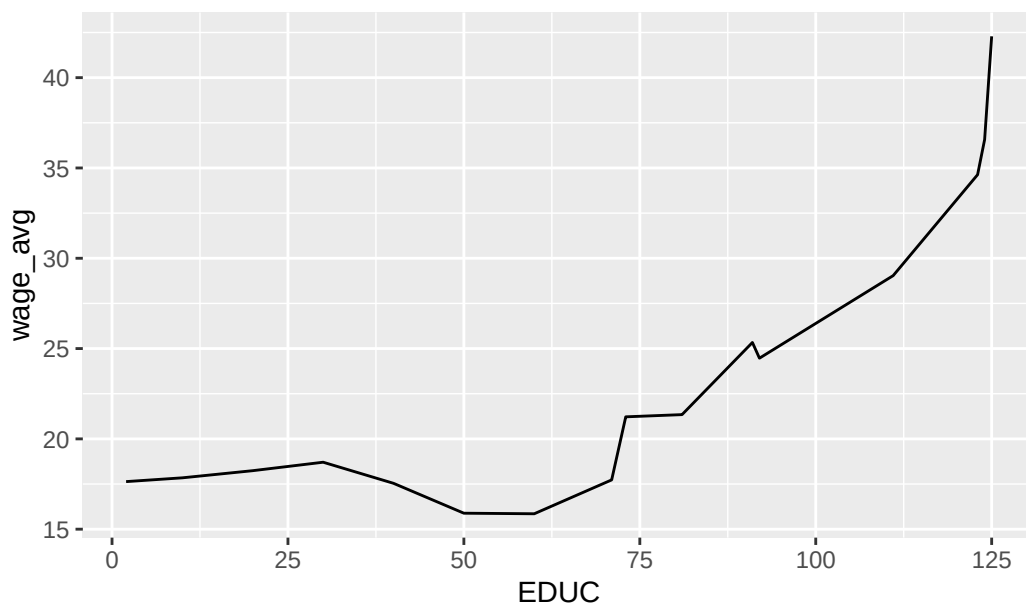
Data Table 1:

```
# A tibble: 2 x 2
  online_work wage_avg
  <lgl>         <dbl>
1 FALSE         22.8
2 TRUE          30.2
```

Graphs:

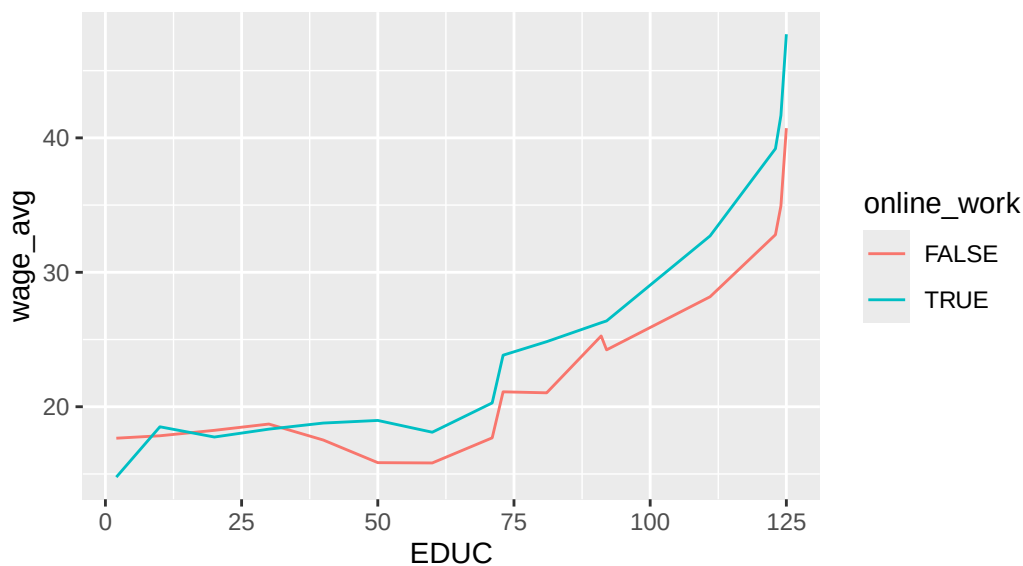
```
# graph of education level to average wage data
data |>
  group_by(EDUC) |>
  summarise(wage_avg= mean(HOURWAGE2)) |>
  ggplot() +
  aes(x= EDUC,y= wage_avg) +
  geom_line() +
  labs(title="Graph 1: Average Wage vs Education Level")
```

Graph 1: Average Wage vs Education Level



```
# graph of average wage of education level of online work vs not-online work
data |>
  mutate(online_work= TELWRKPAY == 1) |>
  group_by(EDUC, online_work) |>
  summarise(count= n(), wage_avg= mean(HOURWAGE2)) |>
  ggplot() +
  aes(x= EDUC,y= wage_avg, color = online_work) +
  geom_line() +
  labs(title="Graph 2: Remote/Non-Remote Average Wage vs Education
          Line")
```

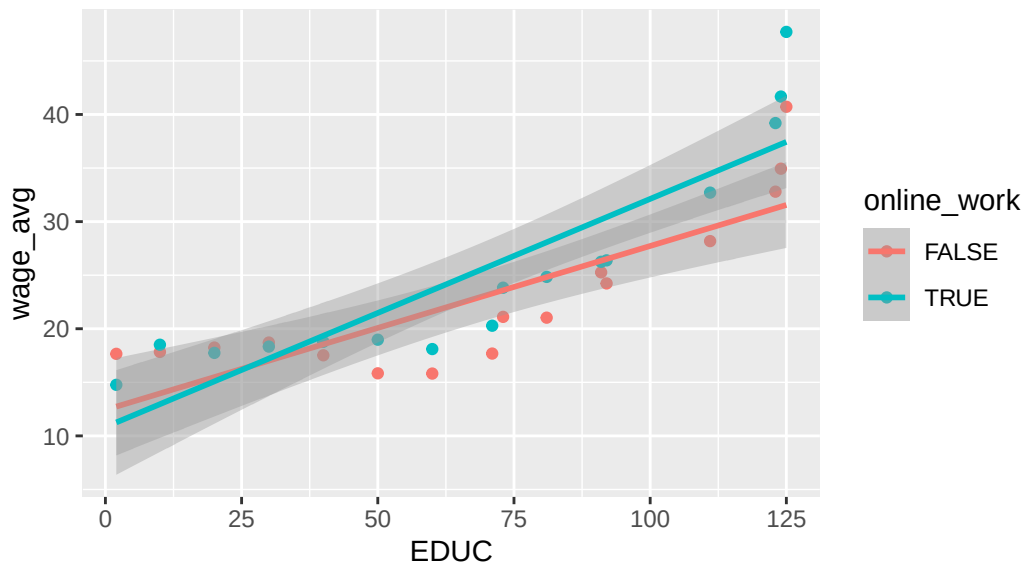
Graph 2: Remote/Non-Remote Average Wage vs Education Line



```
data |>
  mutate(online_work= TELWRKPAY == 1) |>
  group_by(EDUC, online_work) |>
  summarise(count= n(), wage_avg= mean(HOURWAGE2)) |>
  ggplot() +
  aes(x= EDUC,y= wage_avg, color = online_work) +
  geom_point() +
  geom_smooth(method='lm') +
  labs(title="Graph 3: Remote/Non-Remote Average Wage vs Education
          Line Trend ")
```

`geom\_smooth()` using formula = 'y ~ x'

Graph 3: Remote/Non-Remote Average Wage vs Education
Line Trend



```
# graph of average wage over time change of online vs not-online
data |>
  mutate(MONTH = MONTH + (YEAR - 2022) * 12) |>
  mutate(online_work= TELWRKPAY == 1) |>
  group_by(MONTH, online_work) |>
  summarise(wage_avg= (mean(HOURWAGE2))) |>
  ggplot() +
  aes(x= MONTH,y= wage_avg, color= online_work) +
  geom_point() +
  geom_smooth(method='lm') +
  labs(title="Graph 4: Remote/Non-Remote Average Wage vs Monthly
           Growth Trend 2022-2025")
```

`geom\_smooth()` using formula = 'y ~ x'

Graph 4: Remote/Non-Remote Average Wage vs Monthly Growth Trend 2022–2025

